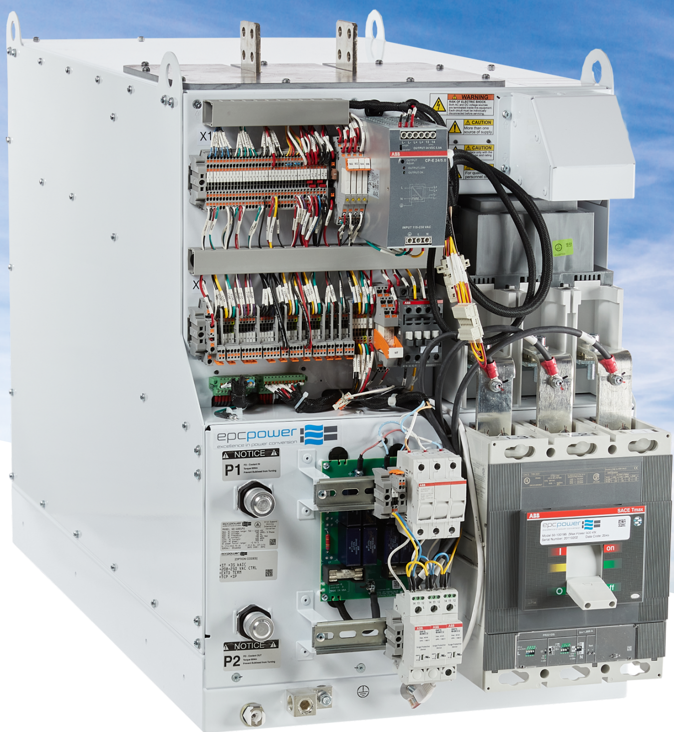
PD250/PD500

250 to 500 kW Energy Storage Inverter

Reliability for the most demanding applications



Engineered to provide years of reliable service in the most demanding applications

EPC's advanced smart inverters for energy storage will enable you to deploy scalable power conversion systems with less effort and less time. Integrating 1,000 V class battery energy storage systems has never been easier or more compact. With world-class power density and an easy to install design, your energy storage system will be up and running in no time.

This inverter is designed from the ground up with simplicity, reliability, and scalability in mind. The liquid-cooled inverter includes an integrated AC contactor, AC breaker, DC contactor, and precharge circuit, enabling simple installation.

Our Power Drawer family provides reliable, abundant power in a small footprint for years of reliable service. Inverters are flexible in AC voltage selection matching with wide range of battery voltage.

Bidirectional Inverter

- THD <2%
- 1250 VDC
- Peak efficiency 98.4%
- 50 & 60 Hz operation
- Grid-tie and off-grid
- Parallel UPS backup
- Real and reactive power control
- Fully bidirectional
- Single-phase capable

Simple O&M

- Easily maintainable
- Modular design with low component count
- Extended warranty available

Return on Investment

- >99% max efficiency
- Low shipping & installation cost
- High dynamic performance

EPC Power Corp. I **epcpower.com**
13250 Gregg Street, Suite A-2 I Poway, CA 92064 I +1.858.748.5590
360 Old Laurens Rd., Suite 800 I Simpsonville, SC 29681 I +1.858.883.9577
info@epcpower.com
November 2023

PD 250 / AC-480 | Model 50-100205
Bidirectional Energy Storage & Microgrid PCS

AC	Nominal AC voltage range (+10%/-12%)	208 VRMS	350 VRMS	400 VRMS	480 VRMS	
	Nominal AC current (export/import)	301 ARMS				
	AC export / import	108 kW	182 kW	208 kW	250 kW	
	Overload capacity	110%, 10 minutes | 125%, 10 seconds				
	AC high voltage ride-through	1.2 pu				
	Nominal frequency range	50 / 60 Hz (configurable)				
	Harmonic distortion	<3% TDDi per IEEE 519				
	Power factor / reactive power	-1 to 1, leading/lagging (full 4-quadrant operation)				
	Maximum aux. power consumption	360 W (control)				
	CEC efficiency (@480 VAC)	98.0%				
	Peak DC to AC efficiency (@480 VAC)	98.4%				
DC	DC voltage range, maximum	310 – 1250 VDC	522 – 1250 VDC	596 – 1250 VDC	715 – 1250 VDC	
	Recommended minimum battery voltage	1.65 x nominal AC voltage				
	Maximum DC current	370 ADC				
	Battery Technology	All battery types, fuel cells, other DC sources, etc.				
	Number of DC inputs	1				
Environmental	Ambient temperature (operation)	-40°C to 50°C				
	Ambient temperature (storage)	-40°C to 50°C				
	Protection degree	IP00 (requires enclosure)				
	Relative humidity	5% - 95% non-condensing				
	Max. elevation	2,000 m [6,500 ft.]				
	Airborne noise	<75 dBA @ 1 m				
	Temperature de-rating	Automatic; see charts				
Cabinet	Maximum dimensions (H x W x D)	mm: [670 x 530 x 1045]	in.: [26.4” x 20.9” x 41.1”]			
	Weight	300 kg [662 lb.]				
	Mounting	Rack mount				
	Cooling	Liquid				
	Cooling Fluid	30% - 50% EGW or PGW				
Certifications	Safety	UL 1741	IEC 62477-1			
	Utility interconnect	UL 1741SA/SB	IEEE 1547	VDE	AS4777.2	
Protections	AC protection	Breaker; 35, 65 kA Isc options				
	DC protection	DC contactor & precharge; ext. fusing required				
	Humidity	By customer				
	Safety features	Overvoltage, overcurrent, overtemperature				
	Ground fault detection	Not included				
Control	Control interface	CAN, Modbus RS485, or Modbus TCP				
	Command latency	1 ms				
	Response time; e.g. step from full charge to discharge	2 ms; adjustable longer via parameters				
	Black-start capable (optional)	Yes; required external control power				
	Grid-tied control modes	Voltage mode	PQ (power)	DQ (current)	cos phi (pf)	STATCOM
	Grid-support functions	Volt/VAR	HzWatt	VoltWatt	Fixed PF	
	Islanded control modes	V&f				
	Control power options	100-120 VAC	208 - 240 VAC	24 VDC		

DC/AC

Ambient temperature (operation) -20°C to 60°C (-40°C with option)

PD 500 / AC-480 | Model 50-100195
Bidirectional Energy Storage & Microgrid PCS

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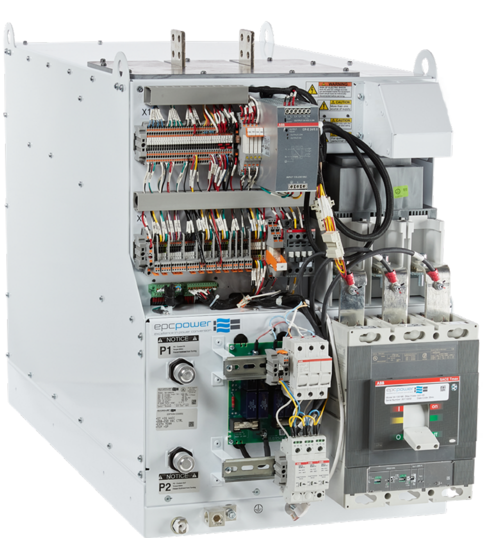
AC	Nominal AC voltage range (+10%/-12%)	208 VRMS	350 VRMS	400 VRMS	480 VRMS	
	Nominal AC current (export/import)	602 ARMS				
	AC export / import	216 kW	365 kW	417 kW	500 kW	
	Overload capacity	110%, 10 minutes	125%, 10 seconds			
	AC high voltage ride-through	1.2 pu				
	Nominal frequency range	50 / 60 Hz (configurable)				
	Harmonic distortion	<3% TDDi per IEEE 519				
	Power factor / reactive power	-1 to 1, leading/lagging (full 4-quadrant operation)				
	Maximum aux. power consumption	1095 W (control)				
	CEC efficiency (@480 VAC)	98.0%				
	Peak DC to AC efficiency (@480 VAC)	98.4%				
DC	DC voltage range, maximum	310 – 1250 VDC	522 – 1250 VDC	596 – 1250 VDC	715 – 1250 VDC	
	Recommended minimum battery voltage	1.65 x nominal AC voltage				
	Maximum DC current	750 VDC				
	Battery Technology	All battery types, fuel cells, other DC sources, etc.				
	Number of DC inputs	1				
Environmental	Ambient temperature (operation)	-40°C to 50°C				
	Ambient temperature (storage)	-40°C to 50°C				
	Protection degree	IP00 (requires enclosure)				
	Relative humidity	5% - 95% non-condensing				
	Max. elevation	2,000 m [6,500 ft.]				
	Airborne noise	<75 dBA @ 1 m				
	Temperature de-rating	Automatic; see charts				
Cabinet	Maximum dimensions (H x W x D)	mm: [670 x 530 x 1045]	in.: [26.4" x 20.9" x 41.1"]			
	Weight	328 kg [725 lb.]				
	Mounting	Rack mount				
	Cooling	Liquid				
	Cooling Fluid	30% - 50% EGW or PGW				
Certifications	Safety	UL 1741:2010 R2.18	IEC 62477-1			
	Utility interconnect	UL 1741SA/SB	IEEE 1547	VDE	AS4777.2	
Protections	AC protection	Breaker; 35, 65 kA Isc options				
	DC protection	DC contactor & precharge; ext. fusing required				
	Humidity	By customer				
	Safety features	Overvoltage, overcurrent, overtemperature				
	Ground fault detection	Not included				
Control	Control interface	CAN, Modbus RS485, or Modbus TCP				
	Command latency	1 ms				
	Response time; e.g. step from full charge to discharge	2ms; adjustable longer via parameters				
	Black-start capable (optional)	Yes; required external control power				
	Grid-tied control modes	Voltage mode	PQ (power)	DQ (current)	cos phi (pf	
	Grid-support functions	Voltage mode	PQ (power)	DQ (current)	cos phi (pf)	STATCOM
	Islanded control modes	V&f				
	Control power options	100-120 VAC	208 - 240 VAC	24 VDC		



DC/AC